Mahmoodur Rahman

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EDUCATION

Degree:

- Master of Data Science — University of British Columbia — 2021-22
- Master of Public Health *in* Epidemiology — Independent University, Bangladesh — 2017-18
- Bachelor of Medicine and Surgery — University of Dhaka — 2005-13

Non-degree credited course:

- Introduction to Computer Programming — Independent University, Bangladesh — 2020
- Calculus and analytical geometry — Independent University, Bangladesh — 2020
- Linear Algebra - vectors and matrices — Independent University, Bangladesh — 2020

Summer School:

- Deep Learning and Reinforcement Learning — Mila - Quebec AI Institute; July 17-21 — 2023

Certificate courses

- Foundations of AI and Machine Learning — AI for Public Health; Oct 20 - Nov 28 — 2025
- Handling Missing Data in Health Research — AI for Public Health; Feb 06 - Mar 13 — 2026
- Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2, 2022)

PROFESSIONAL EXPERIENCE

Provincial Health Services Authority (PHSA), Vancouver BC (cont. 3 years 7 months)

- **Business Analyst, Data and Analytics** **Sep 2023 – Current**
  Stroke Services BC
  - Led the transformation of a SAS-based data science workflow into a reproducible, modular, and version-controlled R-based workflow within an Azure virtual machine, leveraging the Microsoft 365 enterprise ecosystem and Posit Workbench.
  - Conducted statistical analyses to monitor stroke-related performance and operational indicators, including standardized metrics such as the Risk-Adjusted Mortality Rate.
  - Developed an interactive R Shiny dashboard and contributed to the design and implementation of a Power BI dashboard.
  - Analyzed data from the provincial stroke registry and integrated insights from national and provincial administrative datasets, including the Discharge Abstract Database (DAD) and the National Ambulatory Care Reporting System (NACRS).
  - Presented and interpreted findings for diverse stakeholders, including health authorities, the Ministry of Health, software engineers, researchers, and academic partners.
- **Lead, Research and Knowledge Translation** **Jul 2022 – Sep 2023**
  Provincial Infection Control Network (PICNet)
  - Co-led the revision and update of the Provincial Medical Devices Reprocessing SOP.
  - Formulated research questions and conducted critical appraisals of scientific literature to support evidence-based decision-making.
  - Designed and implemented interactive R Shiny dashboards to visualize provincial incidence rates of healthcare-associated infections (HAIs), including Clostridioides difficile (CDI), Carbapenemase-producing organisms (CPO), and Methicillin-resistant Staphylococcus aureus (MRSA), replacing static, non-automated PDF reports.

- Developed interactive R Shiny dashboards using provincial COVID-19 data to present descriptive statistics on child emergency department visits with suspected viral respiratory illness (VRI) symptoms.

Master of Data Science program, University of British Columbia **Apr – Jul 2022**

- Capstone Project: Developed an interactive dashboard to visualize temporal trends in the incidence of chronic diseases using data from the Chronic Disease Registry (CDR) in British Columbia. (*link*)
  - Collaborated with the Office of the Provincial Health Officer and the BC Ministry of Health as key stakeholders.
  - Compared multiple statistical modeling approaches—including a naïve mean-based model, a Bayesian model using Integrated Nested Laplace Approximation (INLA) with Gamma generalized linear smoothing, and Joinpoint regression—to assess and communicate trends over time.
  - Worked in a multidisciplinary team of four, each contributing unique expertise across different components of the project.
  - Tech stack included R, R Shiny, GIS (Leaflet), Bash (Unix shell), Java, and GitHub.

icddr,b, Dhaka, Bangladesh (7 years 4 months)

- **Research Investigator** (Co-Investigator) **Dec 2019 – Jun 2022**
- **Study Physician** **Jan 2018 – Nov 2019**

  Maternal and Child Health Division
  - Led the design and implementation of a nationwide health facility readiness survey using the WHO's Service Availability and Readiness Assessment (SARA) framework. Responsibilities included questionnaire development, study design, sampling strategy, data collection, analysis, reporting, and stakeholder engagement.
  - Developed an automated, R-based interactive HTML report to share findings with stakeholders and donors (e.g., USAID, Save the Children), replacing static Stata-generated Word/PDF reports.
  - Led assessments of Quality of Care (QoC) for maternal, child, adolescent, and newborn health services across health facilities, in collaboration with UNICEF.
  - Contributed to the development of technical reports and peer-reviewed publications.
- **Clinical Fellow** **Aug 2017 – Dec 2017**

  Dhaka Hospital, Nutrition and Clinical Services Division
  - Clinical residency in pediatrics.
- **Study Physician** **Mar 2015 – Dec2016**
- **Project Research Physician** **Aug 2014 – Feb 2015**

  Maternal and Child Health Division
  - Contributed to study design, questionnaire development, data collection management (REDCap), monitoring of research activities, data analysis (Stata and Excel), quality assurance, and manuscript preparation for a study that led to the development of a low-cost jaundice ruler. This project was conducted in collaboration with researchers from Harvard Medical School and Johns Hopkins Bloomberg School of Public Health, funded by the Gates Foundation through the Saving Lives at Birth initiative. (*link*)
  - Participated in clinical research activities, including performing spinal taps and data collection for a related study (*link*)
  - Supported questionnaire development, data collection oversight, data analysis (Stata and Excel), and quality checks (e.g., Modified Ballard Exam) for an NIH-funded (R01)

community-based cluster-randomized controlled trial (*link*). Also contributed to a research project investigating the most accurate biometric parameters in the third trimester for estimating gestational age (*link*).

- Collaborated with multidisciplinary teams including microbiologists, lab technicians, anthropologists, and researchers from Bangladesh, Tanzania, and Pakistan, representing institutions such as WHO, Aga Khan University, Harvard Medical School, Johns Hopkins Bloomberg School of Public Health, and the Child Health Research Foundation (CHRF)

SCHOLARLY CONTRIBUTIONS

Peer-reviewed manuscripts

- Azad R, Billah SM, Bhui BR, Ali NB, Herrera S, de Graft-Johnson J, Garg L, Priyanka SS, Zubair S, Rokonuzzaman SM, **Rahman M**. Mother's care-seeking behavior for neonatal danger signs from qualified providers in rural Bangladesh: A generalized structural equation modeling and mediation analysis. Frontiers in pediatrics. 2023 Jan 4;10:929157.
- Alliance for Maternal and Newborn Health Improvement (AMANHI) GA Study Group (2022). Population-based rates, risk factors and consequences of preterm births in South-Asia and sub-Saharan Africa: A multi-country prospective cohort study. Journal of global health, 12, 04011. https://doi.org/10.7189/jogh.12.04011
- Alliance for Maternal and Newborn Health Improvement (AMANHI) Gestational Age Study Group, Alliance for Maternal and Newborn Health Improvement (AMANHI) GA Study Group. Simplified models to assess newborn gestational age in low-middle income countries: findings from a multicountry, prospective cohort study. BMJ Global Health. 2021 Sep;6(9):e005688.
- Ali NB, Priyanka SS, Bhui BR, Herrera S, Azad MR, Karim A, Shams Z, **Rahman M**, Rokonuzzaman SM, Meena US, El Arifeen S. Prevalence and factors associated with skin-to-skin contact (SSC) practice: findings from a population-based cross-sectional survey in 10 selected districts of Bangladesh. BMC Pregnancy and Childbirth. 2021 Dec;21:1-3.
- Deb S, Mohammed MS, Dhingra U, Dutta A, Ali SM, Dixit P, Juma MH, Hassan MJ, Sazawal S, Nisar I, Ilyas M. Performance of late pregnancy biometry for gestational age dating in low-income and middle-income countries: a prospective, multicountry, population-based cohort study from the WHO Alliance for Maternal and Newborn Health Improvement (AMANHI) Study Group. The Lancet Global Health. 2020 Apr 1;8(4):e545-54.
- Lee AC, Mullany LC, Koffi AK, Rafiqullah I, Khanam R, Folger LV, **Rahman M**, Mitra DK, Labrique A, Christian P, Uddin J. Urinary tract infections in pregnancy in a rural population of Bangladesh: population-based prevalence, risk factors, etiology, and antibiotic resistance. BMC Pregnancy and Childbirth. 2020 Dec;20(1):1-1.
- Schaeffer LE, Ahmed S, **Rahman M,** Whelan R, Rahman S, Roy AD, Nijhum TA, Bably NN, D'Couto H, Hudelson C, Jaben IA. Development and evaluation of a mobile application for case management of small and sick newborns in Bangladesh. BMC medical informatics and decision making. 2019 Dec;19(1):1-2.
- Lee AC, Folger LV, **Rahman M**, Ahmed S, Bably NN, Schaeffer L, Whelan R, Panchal P, Rahman S, Roy AD, Baqui AH. A novel icterometer for hyperbilirubinemia screening in low-resource settings. Pediatrics. 2019 May 1;143(5).
- Lee AC, Mullany LC, Quaiyum M, Mitra DK, Labrique A, Christian P, Ahmed P, Uddin J, Rafiqullah I, DasGupta S, **Rahman M**. Effect of population-based antenatal screening and treatment of genitourinary tract infections on birth outcomes in Sylhet, Bangladesh (MIST): a cluster-randomised clinical trial. The Lancet Global Health. 2019 Jan 1;7(1):e148-59.

- Baqui AH, Lee AC, Koffi AK, Khanam R, Mitra DK, Dasgupta SK, Uddin J, Ahmed P, Rafiqullah I, **Rahman M**, Quaiyum A. Prevalence of and risk factors for abnormal vaginal flora and its association with advese pregnancy outcomes in a rural district in north-east Bangladesh. Acta obstetricia et gynecologica Scandinavica. 2019 Mar;98(3):309-19.
- Baqui A, Ahmed P, Dasgupta SK, Begum N, **Rahman M**, Islam N, Quaiyum M, Kirkwood B, Edmond K, Shannon C, Newton S. Development and validation of a simplified algorithm for neonatal gestational age assessment–protocol for the Alliance for Maternal Newborn Health Improvement (AMANHI) prospective cohort study. Journal of global health. 2017 Dec;7(2).

Evaluation report

- Assessment of the MaMoni Health Systems Strengthening Project, in Bangladesh. *(link)*
- Assessment of licensing status, compliance with the basic licensing requirements, and MNH care readiness of the private inpatient healthcare facilities of Bangladesh
- Findings from the Quality of Care Survey in Manikganj District, MaMoni MNCSP Baseline Survey Report (*link*)

Abstract/ Posters

- Yan T, Heran M, Guest W, Yip A, Cua L, **Rahman M**. Stroke endovascular thrombectomy – time delays of repeated imaging. In: Canadian Association of Radiologists (CAR) Annual Scientific Meeting; 2025 Apr 3–6; Montreal, Canada.
- Evaluation of a Mobile Application for Case Management of Small and Sick Newborns in Bangladesh *(link)*
- Validity of an Icterometer for Neonatal Jaundice Screening in Low-Resource Settings *(link)*
- Lee AC, Folger L, Ahmed S, Schaeffer L, Bably N, **Rahman M**, Whelan R, Panchal P, Roy A, Rahman S, Begum N, Baqui A. Validity of an Icterometer for Jaundice Screening. The Third WHO Global Forum on Medical Devices 2017 May 11 (pp 15).

Blog post

- *Confronting tradition to save newborns*. Field Team Stories, International Center for Maternal and Newborn Health, Johns Hopkins Bloomberg School of Public Health *(link)*

PERSONAL PROJECTS

- Developed a dashboard to visualize the prevalence of confirmed COVID-19 cases, deaths, and recoveries in Bangladesh, Egypt, Pakistan, and Indonesia, providing both country-specific insights and inter-country comparisons. *(link)*
- Group projects during Master of Data Science Program at UBC:
  - Machine learning model to predict toxic mushrooms based on morphology (*link*)
  - Python package to analyze and visualize tweets on user-defined topics (*link*)
  - R package with similar functionality for tweet analysis and visualization (*link*)
  - Designed and deployed a dashboard (Heroku deployment expired) using R Shiny to analyze video game sales in North America (*link*) [*Note: Heroku hosting has expired*]
  - Built and deployed a similar dashboard using Dash (Python) for video game sales analytics. (*link*) [*Note: Heroku hosting has expired*]

AFFILIATIONS

- AI4PH Health Research Training Program: Community Advisory Board member
- UBC Data Science and Health: Cluster member

MENTORING

- UBC Medicine Datathon, UBC Department of Medicine — Mar 22 - 23, 2025
- National Health Datathon 2026 — Mar 7-8, 2026

AWARD

- Global burden of disease (GBD) Technical Workshop (Full funded) Scholarship — 2020
  (Cancelled due to COVID-19 pandemic)